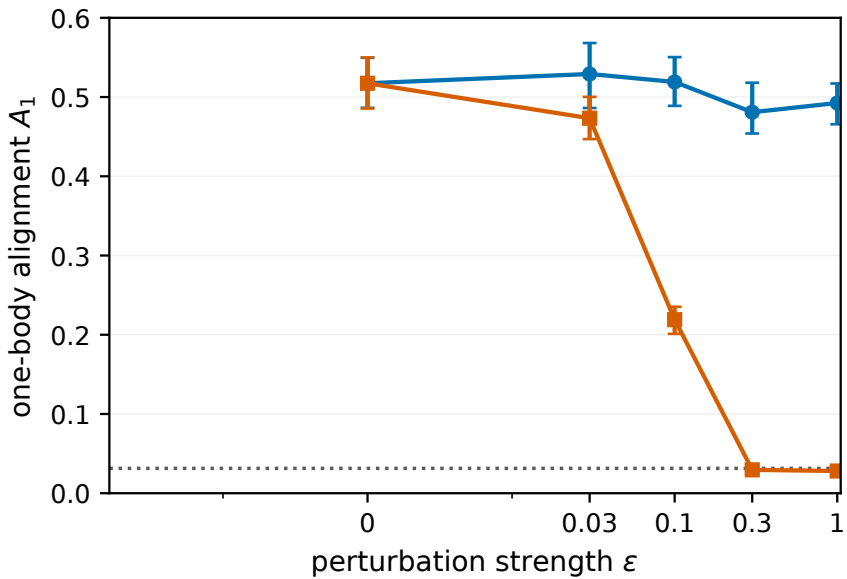




..... rank reference  preserve $U(1)$: R_Z  break $U(1)$: R_X